VACUUM FORMING

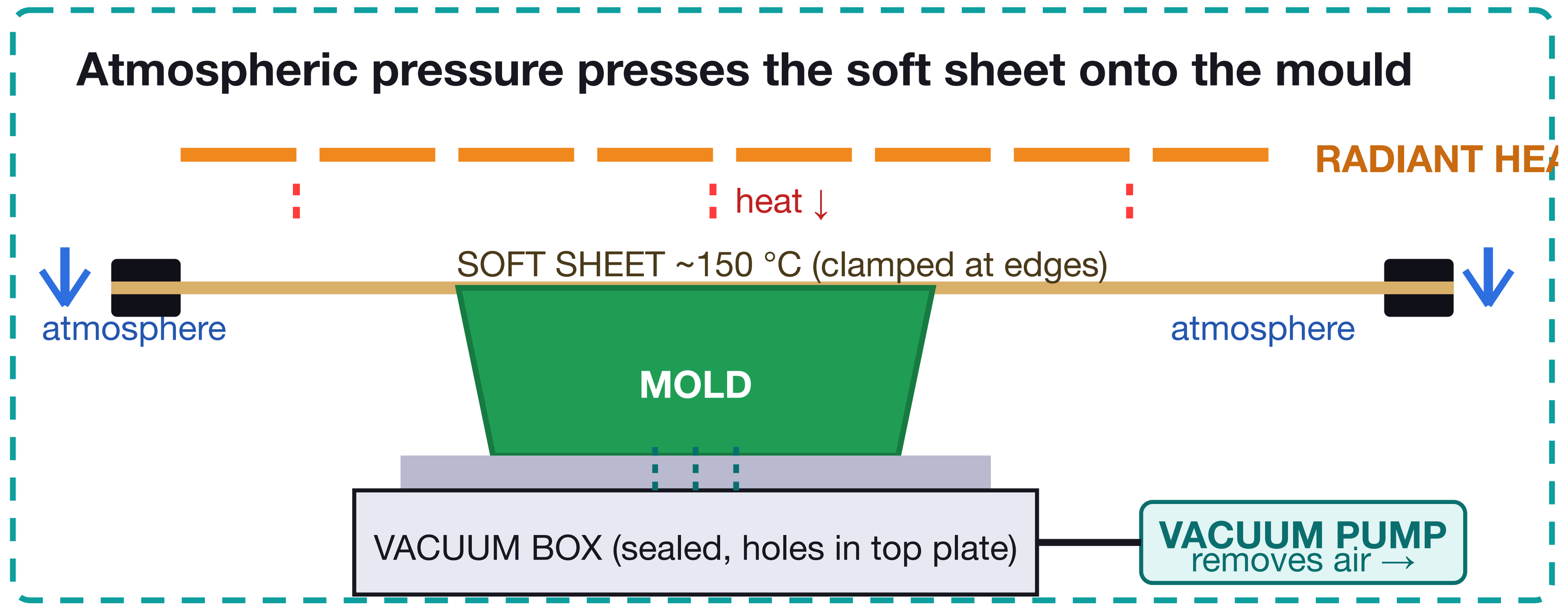
The complete workshop poster — thermoforming physics · machine anatomy · molds · materials · design rules · process · faults · safety · exam facts

140–200 °C	0.1–3 mm	0.7–0.9 bar	1–5 min	hours → days
SHEET FORMING TEMP	SHEET THICKNESS	VACUUM PULL	CYCLE TIME	TOOLING LEAD TIME

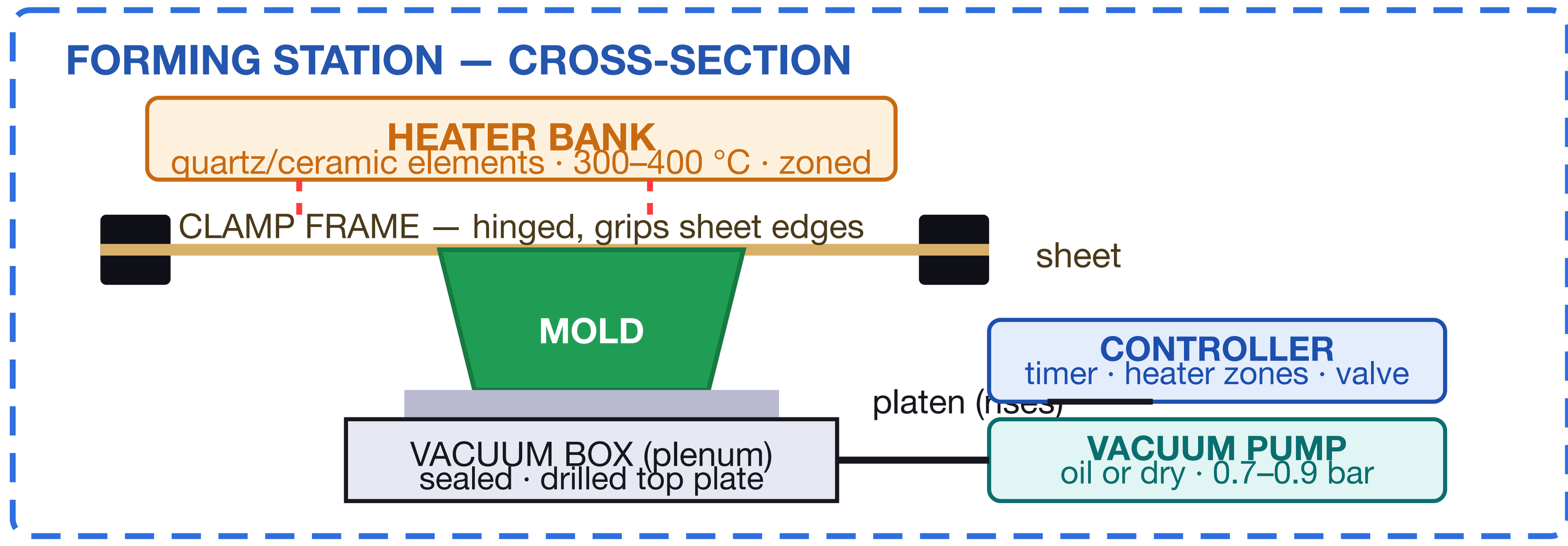
01 HOW VACUUM FORMING WORKS

Vacuum forming is a **thermoforming** process: a flat sheet of **thermoplastic** is heated until soft, then **atmospheric pressure** stretches it over (or into) a mould. The vacuum pump's job is only to **remove the air** — the atmosphere does the pressing.

- **The physics:** sealing the sheet over a box and sucking air out creates a pressure difference. Outside = ~1 bar (101 kPa) of air pushing down; inside ≈ 0.1–0.3 bar. That force, spread over the sheet, presses it into every detail of the mould.
- **Heat first:** the sheet must be above its **glass-transition temperature** (softening point) but below melting — for HIPS ~140–165 °C. Too cold = no detail; too hot = burns, tears, webbing.
- **Thermoplastics only:** the process relies on the sheet re-softening when reheated. **Thermosets** (epoxy, Bakelite) set permanently — they cannot be vacuum formed.
- **Sag tells you it's ready:** as the sheet softens it droops (sags) 10–25 mm under its own weight — the classic "ready to form" signal.



02 MACHINE ANATOMY



- **Heater bank:** radiant elements heat the sheet from above — zoned control (edge vs centre) evens out heat loss at the frame.
- **Clamp frame:** hinged frame holds the sheet; the sheet must overhang the frame on all sides so the seal holds.
- **Platen + vacuum box:** the mould sits on a drilled plate over a sealed chamber; holes/porous mould surface connect the detail to the vacuum.
- **Valve & pump:** a valve releases vacuum to pop the part off; the pump pulls 0.7–0.9 bar. **Reverse-air** valves blow the part free on sticky moulds.

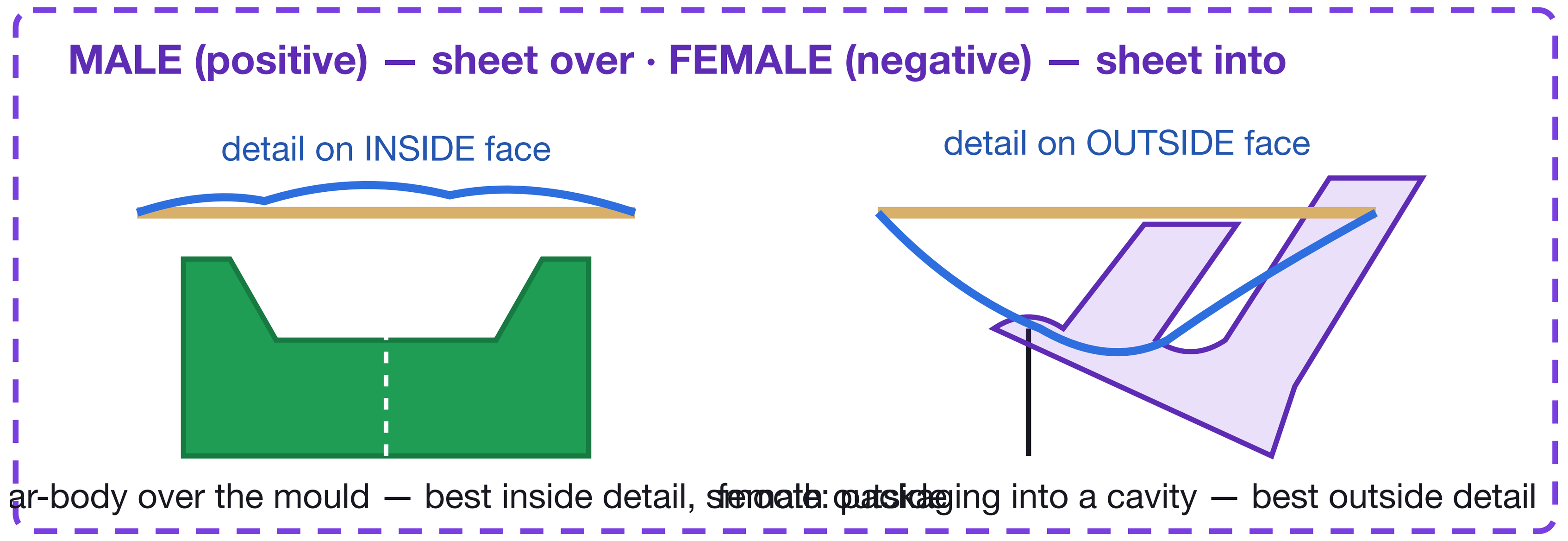
Bench machine: heater on a hinge above, clamp frame, a platen you raise into the sheet, and a small pump underneath. Industrial machines do the same job with bigger heaters and automatic feed.

03 STEP-BY-STEP PROCESS

1. **Design & make the mould** — MDF, plywood or 3D print, with draft angles, radiused edges and vent holes (see panel 07).
2. **Pick the sheet** — material + thickness for the part (0.5–1.5 mm typical); cut slightly larger than the clamp frame opening.
3. **Prep the machine** — clean the mould, sit it on the platen, centre it under the heater, check vent holes are open.
4. **Clamp the sheet** — flat, no wrinkles, evenly gripped all round; check the seal by blocking a port and feeling the pump pull.
5. **Heat** — switch on the heater; wait for the sheet to soften and **sag 10–25 mm** (≈1–3 min for 1 mm HIPS — watch it, don't guess).
6. **Form** — swing the heater away / raise the platen so the mould meets the sagging sheet; stop when it just touches.
7. **Apply vacuum** — open the valve; air evacuates and the atmosphere presses the sheet over the mould (10–30 s).
8. **Cool** — hold vacuum while the sheet sets against the cooler mould; a fan or cool platen speeds it up. Don't rush — soft plastic distorts.
9. **Release** — vent the vacuum (reverse-air helps); the part should pop off the mould cleanly.
10. **Trim** — cut away the clamped waste web with scissors/knife, or trim on the laser/router for precision.
11. **Finish** — sand edges, drill holes, paint or apply graphics, assemble with inserts/adhesive.
12. **Reset** — remove scrap sheet, clean the mould, check for debris before the next cycle.

Cycle timing (1 mm HIPS, bench machine): clamp 15 s · heat 60–120 s · form 10–30 s · cool 30–60 s · trim ~60 s — **total ≈ 3–5 min per part**. Heating is the bottleneck; preheat the mould and use thin sheet to shorten it.

04 MOLD TYPES — MALE vs FEMALE



- **Male (positive) mould:** sheet drapes **over** it — used for trays, covers, model-car bodies. Detail lands on the **inside**; the outside stays smooth. Deeper draws possible, but walls thin at the top corners.
- **Female (negative) mould:** sheet is pulled **into** a cavity — used for packaging, blisters, cases. Detail lands on the **outside**; the inside stays smooth. Best exterior finish for consumer products.
- **Plug assist:** a mechanical plug pushes the soft sheet deep into a female cavity **before** vacuum — evens out wall thickness on deep parts (bucket-like forms).
- **Drape / free forming:** no mould detail — sheet sagged over a simple former or blown into a bubble (dome skylights).
- **Rule of thumb:** male moulds are cheaper and easier to make; female moulds give the finish customers see first.

05 MATERIALS MATRIX — THERMOPLASTIC SHEET

MATERIAL	FORM TEMP	PROPERTIES & TYPICAL USES
HIPS	140–165 °C	Cheap, easy, good detail. **The school/workshop default.** Blister packs, trays, model parts, food punnets.
ABS	150–180 °C	Tough, impact-resistant, nice surface. Casings, RC/model car bodies, luggage, instrument trays.
PETG	150–170 °C	**Clear**, tough, food-safe. Clear packaging, display blisters, point-of-sale stands.
Acrylic (PMMA)	150–180 °C	Clear, rigid, scratches easily, brittle. Signs, displays, machine covers. Must be dry.
PVC	130–160 °C	Cheap, forms easily — but releases **HCl fumes** when hot. Avoid in school; banned on laser cutters.
Polycarbonate	170–190 °C	Very tough, impact + heat resistant. Machine guards, safety shields, aircraft interior panels.
HDPE / PP	130–170 °C	Opaque, tough, chemical-resistant; PP gives living hinges. Harder to form — warps and shrinks unevenly.

- **Only thermoplastics work** — the sheet must soften repeatedly. Thermosets, rubber, paper and foil won't form.
- **Thicker sheet = stiffer part, less detail.** Thin sheet (0.25–0.5 mm) gives crisp detail but fragile walls.
- **Watch moisture:** acrylic, PETG and nylon sheets absorb water — trapped steam causes cloudy blisters. Dry them first.

06 DESIGN RULES FOR GOOD PARTS

- **Draft angles 3–5° minimum** (more on deep moulds) so the part releases from the mould. Vertical walls grip and tear.
- **No undercuts** — the part must lift straight off. Undercuts need split/multi-part moulds; avoid them for one-piece forms.
- **Radius every sharp corner** — internal mould corners ≥ 1 × sheet thickness (2–3 mm). Sharp corners = thin, weak, torn plastic.
- **Depth : width ≤ 1 : 2** for even walls. Deep narrow pockets stretch the sheet until the bottom is paper-thin — use plug assist if you must.
- **Plan for thinning:** forming stretches the sheet, so walls are thinner than the raw sheet. Corners and vertical faces thin most — start with thicker stock.
- **Vent holes:** drill 0.5–1 mm holes in every low spot / pocket so trapped air can escape — otherwise the sheet bridges the detail (blurry corners).
- **Webbing risk in concave corners:** excess sheet folds into valleys. Fix with more draft, cooler sheet, or plug assist (see panel 10).
- **Surface finish transfers:** a glossy polished mould gives a glossy part; sanded/bead-blasted moulds give matte. Texture = feature, use it.
- **Nest parts on the sheet:** leave 25–50 mm between parts and the frame so the vacuum seal and even heating survive.

Design S-L-O-P: **S**imple shapes · **L**ow depth (≤ half width) · **O**pen (no undercuts) · **P**ositive draft. A form that breaks SLOP forms first try.

07 MAKING THE MOLD

MOULD MATERIAL	NOTES
MDF	**Best all-round.** Cheap, easy to machine, naturally porous (helps vacuum), paints/seals well. Sand smooth + seal for glossy parts.
Plywood	Quick to build, but grain shows through on shiny parts; seal edges. Fine for prototypes.
3D printed	Fast iteration, complex shapes easy. Sand + prime to hide layer lines. **PLA deforms above ~60 °C** — use PETG/ABS or keep cycle short & cool.
Plaster / ceramic	Great one-offs, porous (good vacuum), sculptable. Brittle — radius edges, handle gently.
Foam (XPS/PU)	Cheapest, carve/shape fast. Seal with PVA or it melts under hot sheet; keeps the mould cool — actually useful for cooling.
Clay / wax	Sculptable for art forms; must be sealed and supported. Wax melts — keep sheet temps low.

- **Recipe for a good mould:** smooth surface → draft angles → radiused edges → vent holes drilled → sealed (or naturally porous) → screwed/clamped to the platen so it can't move mid-cycle.
- **Vent placement:** 0.5–1 mm holes at the deepest/lowest points — where trapped air would bubble the sheet.

Mould checklist: draft ✓ · vents ✓ · radiused edges ✓ · sealed surface ✓ · secured to platen ✓ — tick all five before the first pull.

08 REAL-WORLD APPLICATIONS

- **Packaging:** blister packs, clamshells, chocolate-box inserts, egg trays, fruit punnets.
- **Medical:** sterile blister trays, surgical instrument organisers, dental models.
- **Automotive:** dashboards, door panels, truck fairings, bumpers (big machines).
- **Transport:** aircraft cabin panels, seat backs, train interior trim (fire-retardant sheet).
- **White goods:** fridge liners, washing-machine tubs, bath & shower trays.
- **Point-of-sale:** sign faces, display trays, protective covers.
- **Hobbies & schools:** model-car bodies, F1-in-Schools shells, masks, costume armour, moulds for casting.
- **Prototyping:** quick visual/ergonomic models before injection moulding.

SCALE	WHAT'S TYPICAL
One-off / prototype	School projects, product development — mould from MDF or 3D print, one sheet at a time.
Short run (100s–1000s)	POS displays, custom packaging, jigs & trays — vacuum forming is the economic sweet spot.
Mass production (millions)	Blister packs on roll-fed automatic machines — sheets fed from a reel, formed, trimmed inline.

Why it's everywhere: tooling costs almost nothing compared with injection moulding, so **short runs and prototypes are economic** — yet the same machines scale to millions of blister packs a year.

09 ADVANTAGES vs DISADVANTAGES

✓ ADVANTAGES

- Very cheap tooling — hours/days, not weeks/months.
- Fast setup; low-volume and prototype runs are economic.
- Thin, light parts; large sheets → big parts (bath trays).
- Colour and texture are built into the sheet.
- Smooth, seamless surfaces — no parting lines.
- Same machine makes prototypes and production parts.
- Simple, safe process — ideal for schools.

✗ DISADVANTAGES

- Thermoplastics only — no thermosets, no metals.
- Uneven wall thickness — thin corners, thick flats.
- Limited detail: no sharp corners, no undercuts.
- Webbing in concave features; trim waste (the web).
- Detail on one side only.
- Low dimensional accuracy (±0.5–1 mm) vs injection moulding.
- Deep draws need plug assist or become paper-thin.

Exam comparison: vs injection moulding — cheaper tooling & faster to market, but thicker/uneven walls, lower detail, and higher per-part cost at volume.

10 FAULTS & FIXES

SYMPTOM	CAUSE	FIX
Webbing / folds in corners	Sheet too hot · too much vacuum · concave detail	Cool sheet, reduce heat time, add draft, plug assist
Thin or torn corners	Sharp mould edges · too deep · sheet too hot	Radius corners, thicker sheet, less heat
Poor detail / blurry features	Not hot enough · weak vacuum · blocked vents	More heat, check pump & seal, clear vent holes
Burn marks / bubbles	Overheated or left too long under heater	Shorten heat time, lower heater power
Part sticks to mould	No draft · mould too rough/hot · vacuum not released	Add draft, smooth + seal mould, use reverse-air valve
Uneven thickness across part	Uneven heating (frame heat-sink) · sheet off-centre	Zone the heater, centre sheet, wait for even sag
White stress marks (HIPS)	Stretched too far while too cold	More heat before forming, shallower draw
Wrinkles at sheet edge	Sheet over-clamped / too big for frame	Trim sheet, clamp evenly with moderate force
PLA mould deforms	Mould too hot during cycle	Use PETG/ABS mould or shorter, cooler cycles

Golden rule: 90 % of faults trace back to **heat** (too hot/cold/uneven), **draft** (missing), or **vacuum** (leaks/blocked vents). Fix those three before anything else.

11 SAFETY — READ THIS FIRST

- HOT The heater bank runs at **300–400 °C** and the sheet reaches **140–200 °C** — both cause instant deep burns. Never touch the sheet while it's soft; it sticks to skin.
- GLOVES Wear heat-resistant gloves when handling the clamp frame, mould and fresh parts. The frame radiates heat even after the heater swings away.
- FUMES Hot plastic releases fumes — styrene from HIPS, **HCl gas from PVC** (avoid entirely). Run the extractor fan / vent the room; keep cycle times short.
- COOLING Wait for the part to set fully before releasing — soft plastic distorts, and warm parts still burn. A fan speeds cooling safely.
- PINCH The clamp frame and rising platen are **pinch points** — keep fingers clear of the mechanism, not just the heater.
- UNATTENDED Never leave a machine heating unattended — a forgotten sheet can overheat, ignite and drip burning plastic.
- FIRE Keep an extinguisher nearby; if the sheet ignites, kill the heater first, then extinguish. Never pour water on hot plastic/oil.
- TRIMMING Freshly trimmed edges are **sharp** — use a knife with a proper cutting mat and cut away from your body.

Before you start: extractor on ✓ · gloves on ✓ · extinguisher nearby ✓ · mould secured ✓ · heater timer set ✓. Treat the machine with respect and it's one of the safest workshop processes there is.

12 GLOSSARY + EXAM FACTS (IGCSE DT)

- **Thermoforming:** shaping heated thermoplastic sheet with pressure/vacuum.
- **Thermoplastic:** softens when heated, sets when cool — can be reformed.
- **Thermoset:** sets permanently after first cure — cannot be reformed.
- **Glass-transition temp (Tg):** the temperature where the sheet softens enough to form.
- **Draft angle:** taper on mould walls (3–5°) so the part releases.
- **Undercut:** any feature that locks the part onto the mould.
- **Webbing:** excess-sheet folds in concave corners.
- **Plug assist:** mechanical pusher for deep, even-walled parts.
- **Platen / plenum:** the rising table and its sealed vacuum chamber.
- **Blister / clamshell:** packaging formed over or around the product.

Top 6 exam points (0445):

- Heat softens the sheet; **atmospheric pressure** (not the pump) does the forming.
- Only **thermoplastics** can be vacuum formed.
- Male mould → detail **inside**; female → detail **outside**.
- Design needs **draft, no undercuts, radiused corners**.
- Cheapest tooling of the mass-production processes — ideal for short runs.
- Blister packaging is the most common application.

Compare with: injection moulding (expensive tooling, high detail, high volume) · blow moulding (hollow bottles) · line bending (straight folds only).